

KARGASOK, Russian Federation

WMO: 291220

Lat: **59.0651N**Lon: **80.8325E**Elev: **57**StdP: **100.64**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.2	-37.9	-43.4	0.1	-40.7	-40.3	0.1	-37.4	7.8	-11.1	7.2	-11.5	1.1	270	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	28.0	19.1	26.4	18.5	24.7	17.7	20.8	25.6	19.7	24.4	18.7	23.2	2.6	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.0	23.3	17.9	13.0	22.1	16.8	12.0	21.0	60.4	25.6	56.6	24.6	53.3	23.2	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.7	6.0	DB	-42.9	30.9	4.0	1.5	-45.8	32.0	-48.1	32.9	-50.4	33.7	-53.3	34.8	
(5)			WB	-42.6	22.3	3.5	1.3	-45.2	23.2	-47.3	24.0	-49.3	24.7	-51.8	25.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-21.0	-17.3	-7.7	0.1	8.2	16.3	18.4	14.8	8.6	0.7	-10.9	-17.4	(6)
(7)		DBStd	15.10	9.58	8.53	7.06	6.38	5.91	4.60	3.45	3.68	4.54	5.40	9.11	9.56	(7)
(8)		HDD10.0	4537	961	763	549	298	106	7	0	4	81	291	628	850	(8)
(9)		HDD18.3	6967	1219	997	807	546	317	91	43	121	294	547	878	1108	(9)
(10)		CDD10.0	700	0	0	0	2	52	194	261	153	37	2	0	0	(10)
(11)		CDD18.3	90	0	0	0	0	3	29	46	11	1	0	0	0	(11)
(12)		CDH23.3	671	0	0	0	1	41	233	309	79	8	0	0	0	(12)
(13)		CDH26.7	119	0	0	0	0	8	49	52	9	1	0	0	0	(13)
(14)	Wind	WSAvg	2.9	2.6	2.9	3.2	3.3	3.4	2.6	2.3	2.4	2.7	3.2	3.1	2.9	(14)
(15)	Precipitation	PrecAvg	483	18	15	18	22	46	67	75	67	50	41	36	28	(15)
(16)		PrecMax	682	50	36	46	52	121	145	138	138	107	84	59	46	(16)
(17)		PrecMin	356	3	1	2	2	12	11	23	3	4	17	15	8	(17)
(18)		PrecStd	84	11	8	11	10	26	30	32	31	21	15	12	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	0.9	7.3	18.0	27.7	30.2	30.3	28.0	24.3	15.6	5.1	0.5	(19)
(20)			MCWB	-1.0	-0.5	2.9	9.5	16.1	18.8	21.3	19.5	15.4	10.1	3.6	0.0	(20)
(21)		2%	DB	-2.9	-1.2	4.7	14.1	24.0	28.0	28.0	25.5	20.8	12.4	2.8	-1.1	(21)
(22)			MCWB	-3.5	-2.2	1.0	8.0	14.3	18.7	19.7	18.5	14.9	8.4	1.7	-1.7	(22)
(23)		5%	DB	-5.5	-3.1	2.8	11.3	20.8	26.1	26.6	23.6	18.3	10.2	1.3	-2.7	(23)
(24)			MCWB	-6.2	-4.1	0.2	6.4	12.9	17.9	19.3	17.5	13.3	7.5	0.6	-3.4	(24)
(25)		10%	DB	-7.9	-5.7	1.3	9.0	17.7	24.0	25.0	21.6	15.8	8.1	0.2	-4.9	(25)
(26)			MCWB	-8.5	-6.7	-0.5	4.9	11.5	17.1	18.5	16.7	12.0	6.0	-0.5	-5.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.9	-0.1	3.4	10.7	17.0	21.8	22.7	20.9	17.5	10.9	3.7	0.0	(27)
(28)			MCD B	-0.4	0.8	6.7	16.4	25.5	26.9	27.0	25.7	21.7	13.7	4.7	0.3	(28)
(29)		2%	WB	-3.5	-2.1	1.6	8.5	15.1	20.0	21.4	19.5	15.4	9.1	1.9	-1.6	(29)
(30)			MCD B	-3.0	-1.2	3.6	13.2	22.1	25.3	26.2	24.0	19.8	11.8	2.5	-1.1	(30)
(31)		5%	WB	-6.1	-4.1	0.6	6.8	13.6	18.9	20.2	18.2	13.7	7.6	0.7	-3.3	(31)
(32)			MCD B	-5.6	-3.2	2.4	11.1	19.6	24.4	24.9	22.3	17.4	10.0	1.3	-2.8	(32)
(33)		10%	WB	-8.4	-6.7	-0.5	5.0	12.1	17.7	19.2	17.1	12.3	5.9	-0.4	-5.4	(33)
(34)			MCD B	-7.9	-5.7	1.4	8.7	17.3	22.9	23.6	20.8	15.4	7.9	0.2	-4.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.5	8.3	9.6	9.4	9.0	9.0	8.8	7.8	5.6	6.2	7.2	(35)	
(36)			MCDBR	7.1	6.8	9.4	12.0	13.8	12.6	11.5	12.0	12.1	8.4	5.0	7.6	(36)
(37)			MCWBR	6.9	6.2	6.9	7.2	6.6	5.5	5.2	6.1	6.7	5.3	4.9	7.5	(37)
(38)		5% WB	MCD BR	7.0	6.7	8.3	11.5	12.4	10.9	10.2	10.4	10.9	7.9	5.2	7.5	(38)
(39)			MCWBR	6.8	6.2	6.5	7.3	6.5	5.3	5.2	5.8	6.4	5.3	5.2	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.273	0.308	0.359	0.385	0.393	0.406	0.443	0.411	0.363	0.334	0.289	0.270	(40)
(41)		taud		1.982	2.003	1.946	2.072	2.276	2.328	2.196	2.269	2.361	2.260	2.033	1.926	(41)
(42)		Ebn at Noon		487	670	744	807	835	829	782	779	762	654	461	339	(42)
(43)		Edh at Noon		54	90	129	136	119	115	129	112	87	70	51	40	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.56	1.51	3.05	4.46	4.94	5.70	5.14	3.83	2.42	1.16	0.51	0.31	(44)
(45)		RadStd		0.05	0.11	0.19	0.40	0.56	0.57	0.51	0.44	0.32	0.11	0.11	0.03	(45)